ONTAP Essentials

Cloud Insights

NetApp
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ONTAP Essentials

ONTAP Essentials is a set of dashboards and workflows that provide detailed overviews of your ONTAP inventories and workloads. You may see the following terms used when working in ONTAP Essentials:

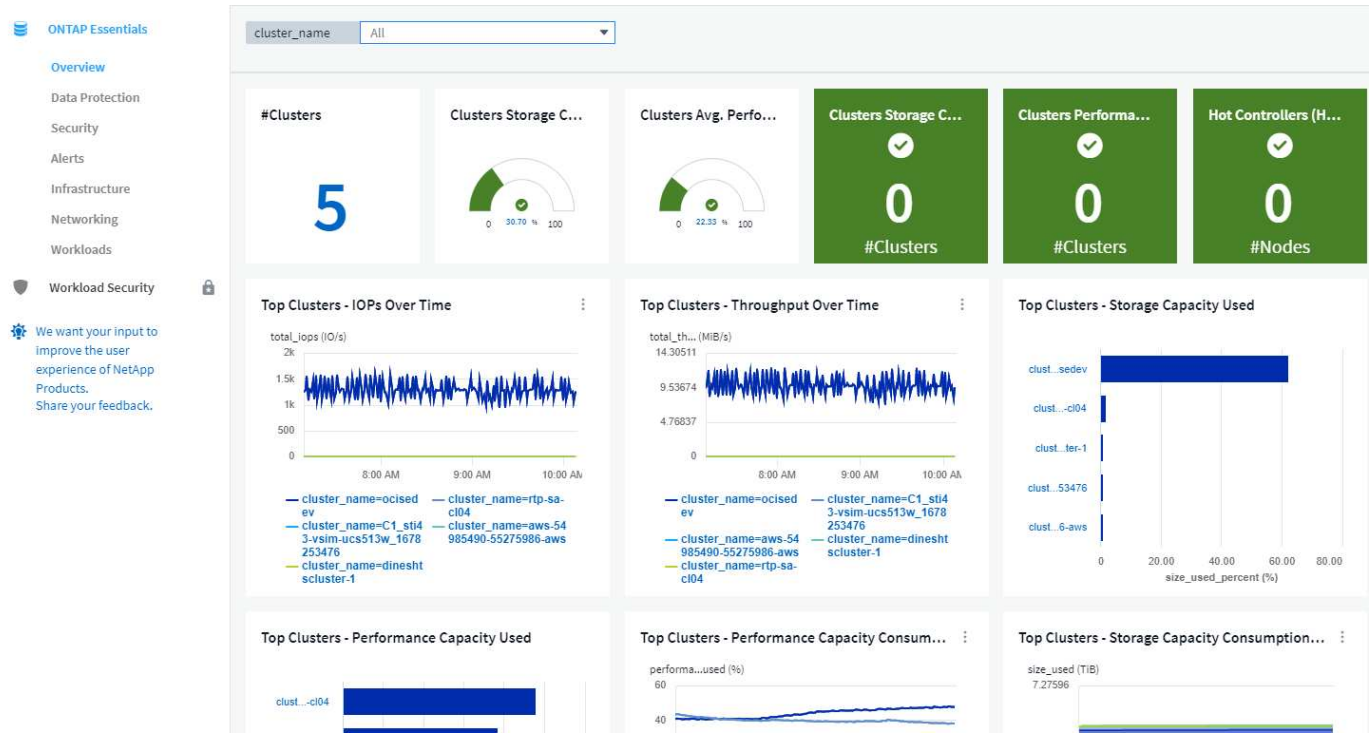
- Infrastructure/Inventory: Objects that provide storage/networking services to user data
- Workloads: Objects that provide interface to users to read/write data.
- Data Protection: Objects that can be protected using NetApp data protection technologies

For additional terms and definitions related to ONTAP, see the ONTAP Data Collector documentation.

ONTAP Essentials requires at least one working ONTAP data collector with data collected within the last seven days.

Overview

To begin exploring, select **ONTAP Essentials** from the main Cloud Insights menu.



The **Overview** dashboard displays useful information like the number of clusters in your environment with their overall capacity and performance percentages. You will also see predictive data regarding the number of expected days until storage capacity or performance capacity runs out of space. Additionally, if any controllers in your infrastructure are running with their CPU at more than 65%--potentially putting your cluster at risk in case of failover—ONTAP Essentials shows those as "Hot" controllers.

Informative graphs give you a look into performance over time as well as breakdowns of capacity usage. Each of these graphs or data points can be used as a starting point for exploration or investigation.

Note: A "days to full" number of "0" (zero) indicates that days to full is estimated at greater than 90 days. In other words, your systems aren’t in danger of running out of space any time soon.

Data Protection

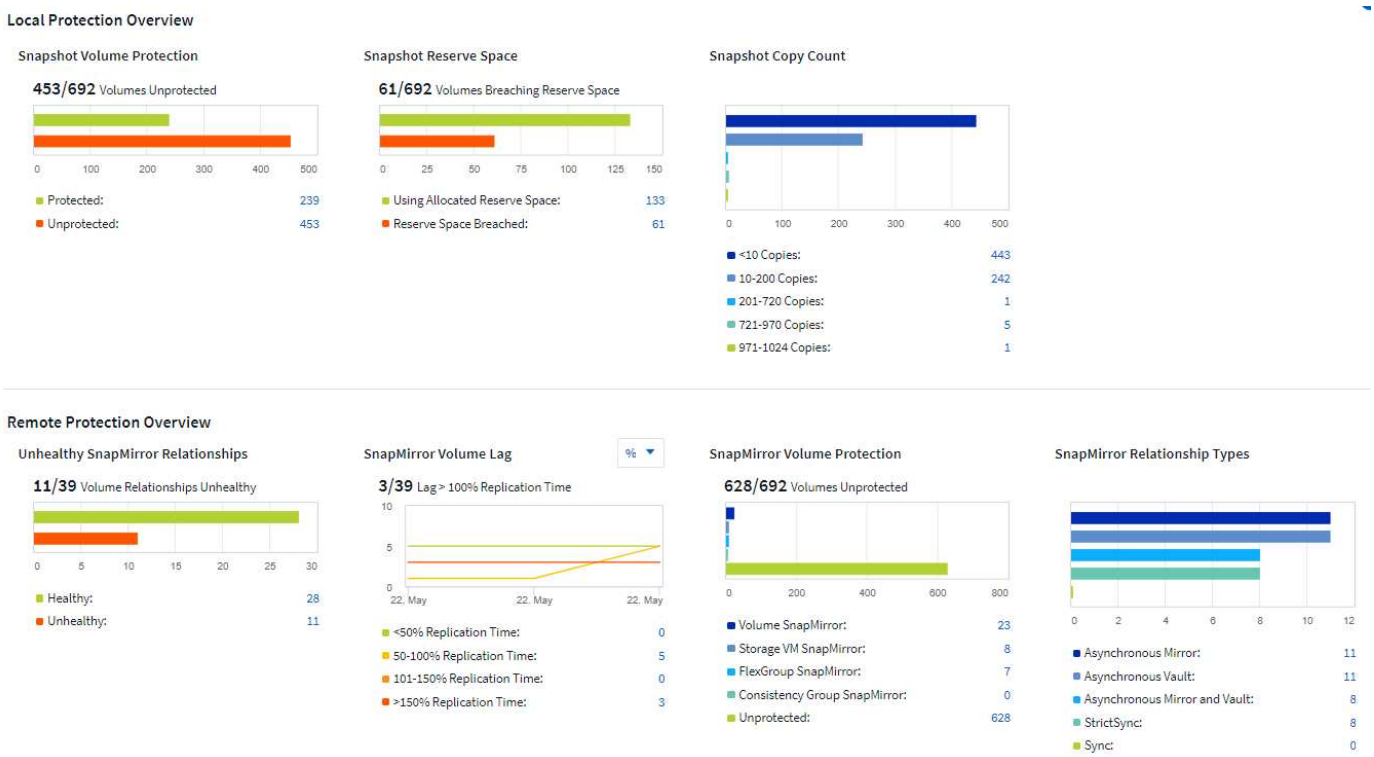
The **Data Protection** page shows the status of volumes protected by **Snapshot copies** or **SnapMirror policies**.

In the *Local Protection Overview* section, the charts provide the following information for volumes protected by Snapshot copies:

- The number of volumes protected by Snapshot copies, as well as those not protected.
- The number of volumes that are using or exceeding the reserve space for the Snapshot copies.
- The number of volumes in specific ranges of Snapshot copy count (i.e. less than 10 copies, 10 to 200, etc.).

In the *Remote Protection Overview* section, the charts provide information related to volumes protected by SnapMirror policies:

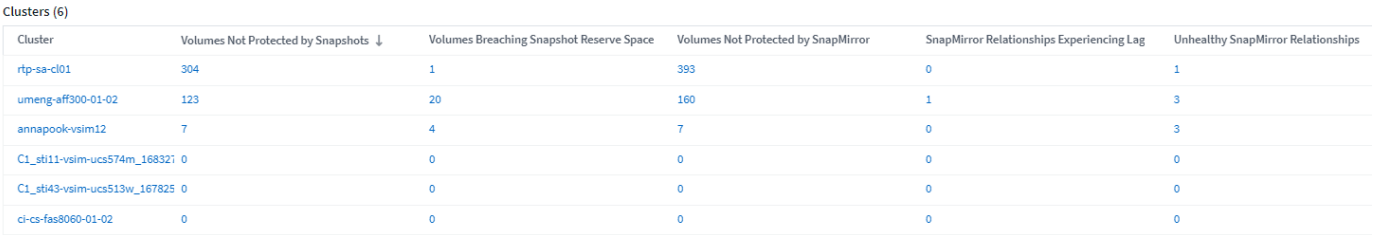
- The number of healthy and unhealthy SnapMirror relationships.
- The number of SnapMirror relationships experiencing recovery point objective (RPO) lag based on the lag status.
- The number of relationships protected by SnapMirror volume protection types such as Volume SnapMirror, SVMDR relationships, FlexGroup SnapMirror relationships, SnapMirror Business Continuity (SMBC) consistency Group relationships, as well as unprotected volumes.
- The number of relationships protected by the SnapMirror relationship types such as Asynchronous Mirror, Asynchronous Vault, Asynchronous MirrorVault, StrictSync, and Sync.



The *Clusters* grid at the bottom of the page provides details related to the following:

- Volumes not protected by Snapshots.

- Volumes breaching snapshot reserve space.
- Volumes not protected by snapmirror policies and snapmirror relationships experiencing lag.
- Unhealthy SnapMirror relationships.



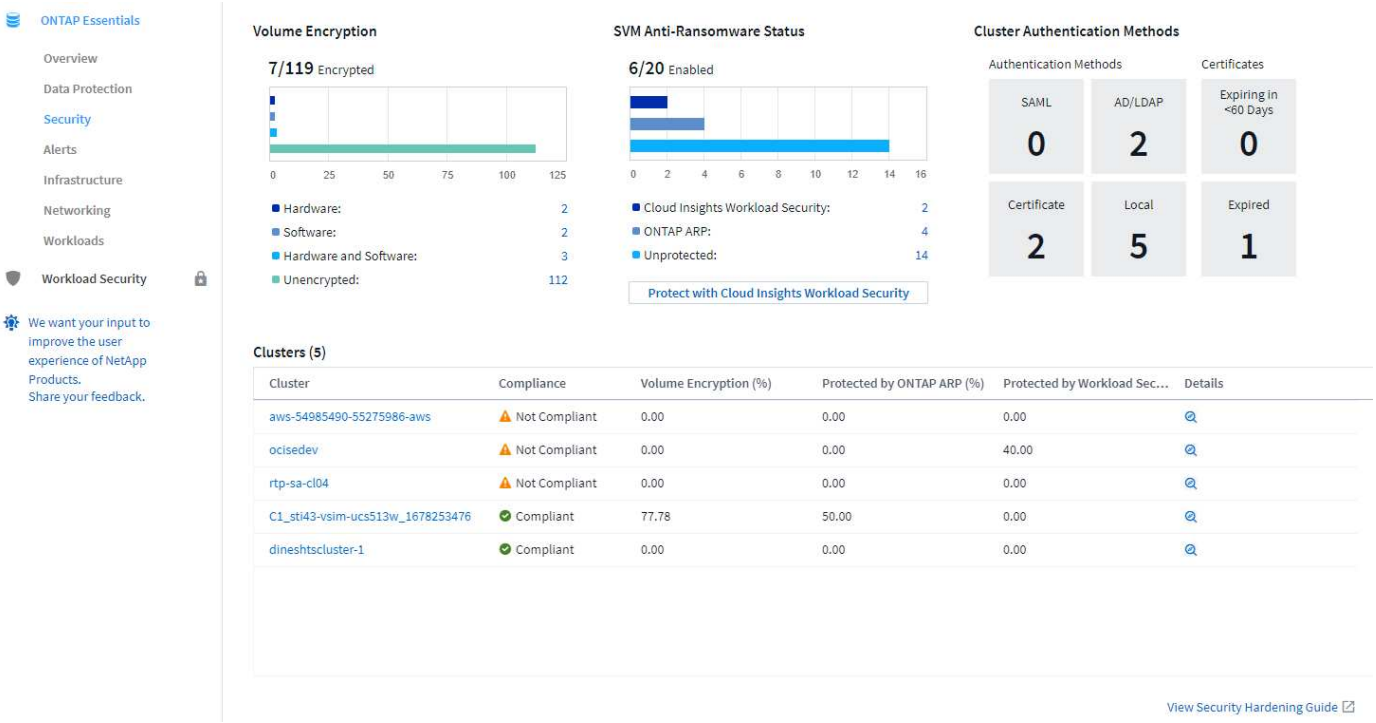
Clusters (6)

Cluster	Volumes Not Protected by Snapshots ↓	Volumes Breaching Snapshot Reserve Space	Volumes Not Protected by SnapMirror	SnapMirror Relationships Experiencing Lag	Unhealthy SnapMirror Relationships
rtp-sa-cl01	304	1	393	0	1
umeng-aff300-01-02	123	20	160	1	3
annapook-vsim12	7	4	7	0	3
C1_sti11-vsim-ucs574m_168327	0	0	0	0	0
C1_sti43-vsim-ucs513w_167825	0	0	0	0	0
ci-cs-fas8060-01-02	0	0	0	0	0

Security

The Security Dashboard gives you an instant view of your current security situation, showing charts for hardware and software volume encryption, anti-ransomware status, and cluster authentication methods. Security criteria is evaluated based on recommendations defined in the **NetApp Security Hardening Guide for ONTAP 9**.

Select any of the encryption or anti-ransomware counts to dive into your environment.

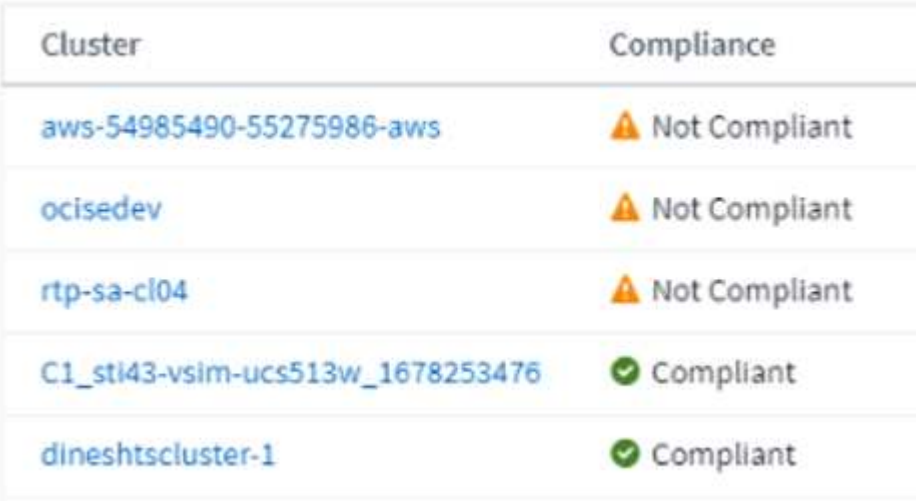


The ONTAP Essentials Security dashboard monitors your environment to determine cluster compliance status. Refer to the Cluster Compliance Categories to learn more. ONTAP Essentials uses the following monitors to determine compliance:

Monitor Name	Attribute Name (Displayed in Cluster Details)	Attribute Compliant Value
FIPS Mode Disabled	FIPS mode	Enabled
Cluster Insecure ciphers for SSH	Secure SSH Settings	Yes

Monitor Name	Attribute Name (Displayed in Cluster Details)	Attribute Compliant Value
Telnet Protocol Enabled	Telnet	Disabled
Remote Shell Enabled	Remote Shell	Disabled
Default Local Admin User Enabled	Default Admin User	Disabled
MD5 Hashed password	MD5 in use	No
Cluster Peer Communication Not Encrypted	Cluster Peering	Encrypted/ No Peer
AutoSupport HTTPS Transport Disabled	AutoSupport using HTTPS	Yes
No NTP Servers are Configured	Network Time Protocol	Configured
NTP Server Count is Low	Network Time Protocol	Configured
Cluster Login Banner Disabled	Login Banner	Enabled
Log Forwarding Not Encrypted	Log Forwarding Encrypted	Yes

Note that if a monitor above is disabled, the cluster details will show the value as ‘Not checked’ for the corresponding security compliance attribute.

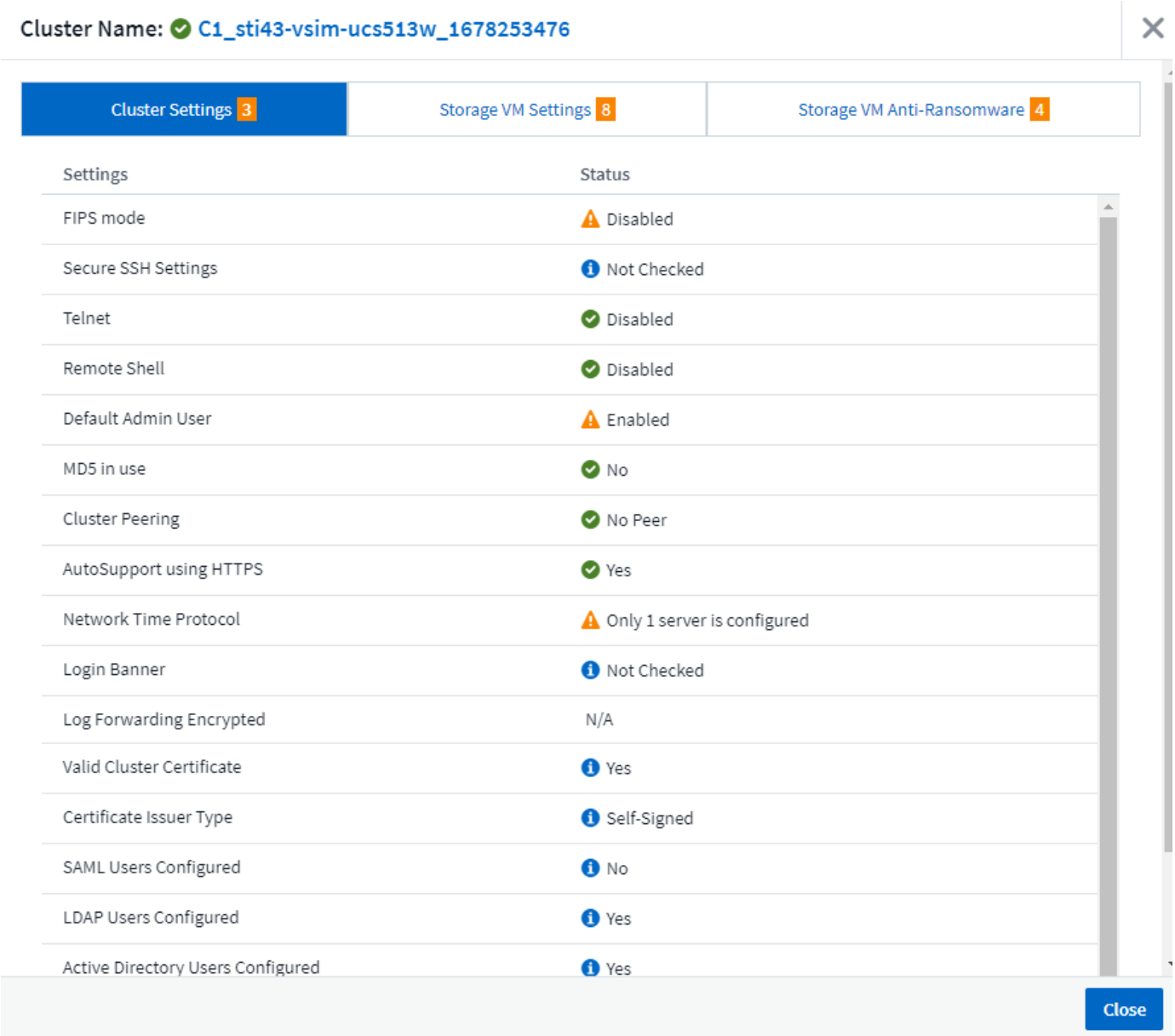


For SVMs, the Security dashboard looks at the following monitors:

Monitor Name	Attribute Name (Displayed in Storage VM Settings)	Attribute Compliant Value
Storage VM Insecure ciphers for SSH	Secure SSH Settings	Yes
Storage VM Login banner disabled	Login Banner	Enabled
Storage VM Audit Log Disabled	Audit Log	Enabled

In the cluster list, select *View Details* for each cluster to open a "slideout" panel showing you the current settings for *Cluster, Storage VM,* or *Anti-Ransomware*.

Cluster details include connection status, certificate information, and more:



Storage VM details show audit and SSH information:

Cluster Name: rtp-sa-cl04

Cluster Settings 2 | Storage VM Settings 7 | Storage VM Anti-Ransomware 8

Storage VM	Login Banner	Audit Log	Secure SSH Settings
mattsvm07_04	Disabled	N/A	Yes
sf-svmdr1	Disabled	N/A	Yes
ss_balajicifs	Disabled	N/A	Yes
ss_balajicifs_1_encrypted	Disabled	N/A	Yes
test1	Enabled	Disabled	Yes
test2	Disabled	N/A	Yes
test3	Disabled	N/A	Yes
cl04_data_svm1	Enabled	Enabled	Yes

Anti-Ransomware details show whether a storage VM is protected by ONTAP's Anti-Ransomware Protection or Cloud Insights Workload Security. Note that the ONTAP ARP column displays the current status of ONTAP's on-board Anti-Ransomware Protection, which is configured on the ONTAP system. Cloud Insights Workload Security can be enabled by selecting "Protect" in that column.

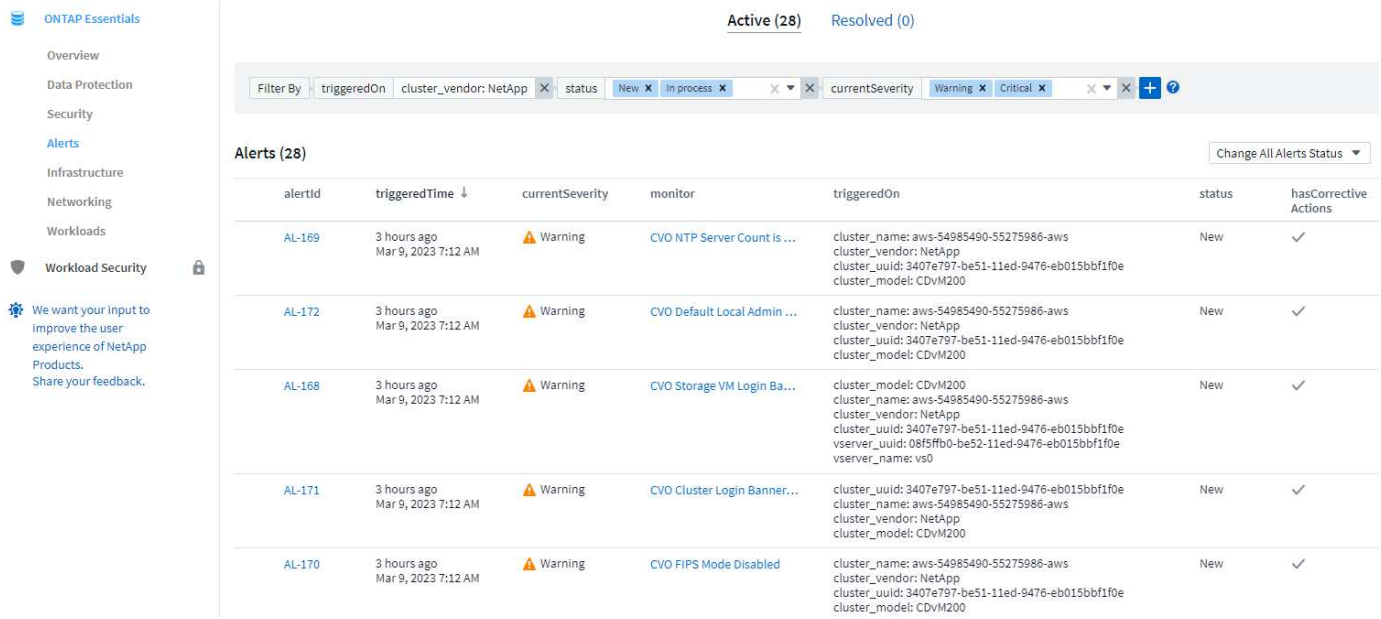
Cluster Name: ocisedev

Cluster Settings 7 | Storage VM Settings 5 | Storage VM Anti-Ransomware 3

Storage VM	Protected by Workload Security	Protected by ONTAP ARP
CloudComplianceSVM	Protect	N/A
t1appSVM01	Protect	N/A
tawny_mirror	Protect	N/A
demoGroupShares	Protected	N/A
demoGroupShares2	Protected	N/A

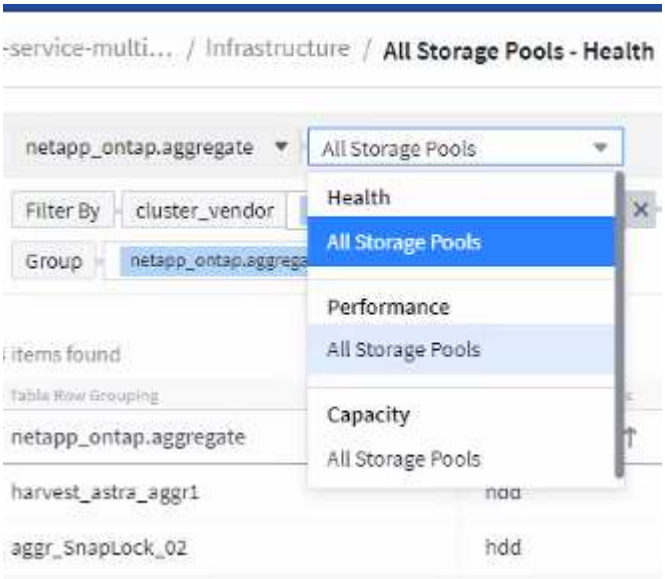
Alerts

Here you can view the Active alerts in your environment and quickly drill down into potential problems. Select the *Resolved* tab to view alerts that have been resolved.

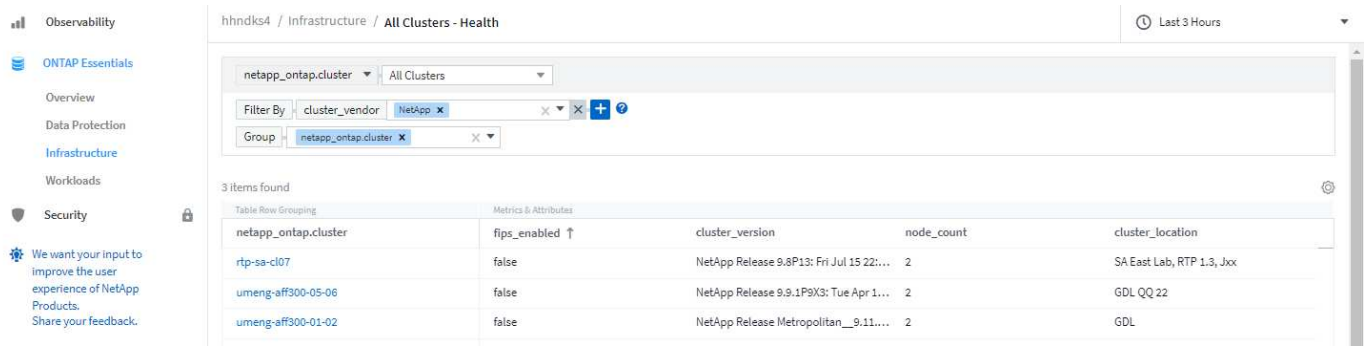


Infrastructure

The ONTAP Essentials **Infrastructure** page gives you a view of cluster health and performance, using pre-built (yet further customizable) queries on all the basic ONTAP objects. Select the object type you wish to explore (cluster, storage pool, etc.) and choose whether to view health or performance information. Set filters to dive deeper into individual systems.

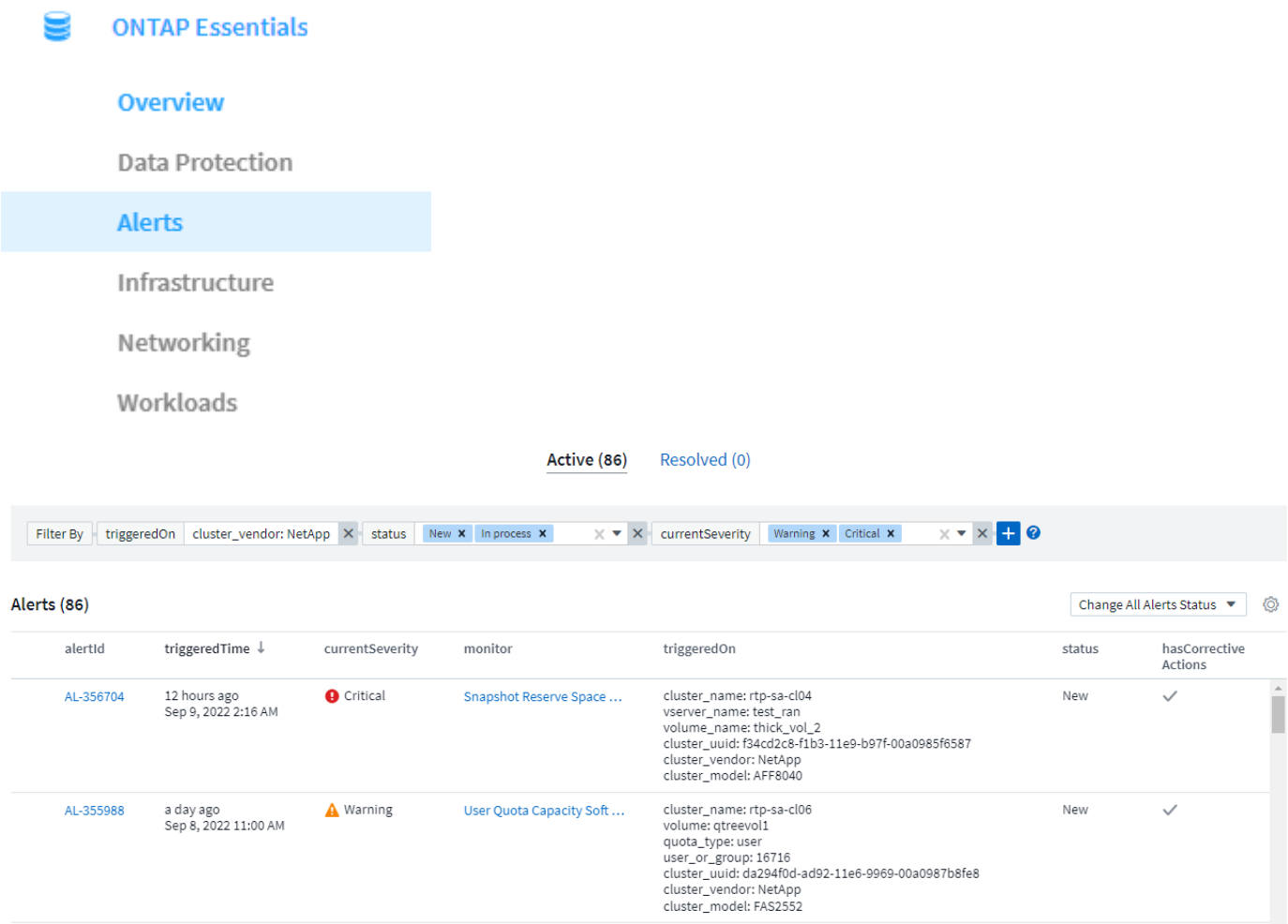


Infrastructure page showing cluster health:



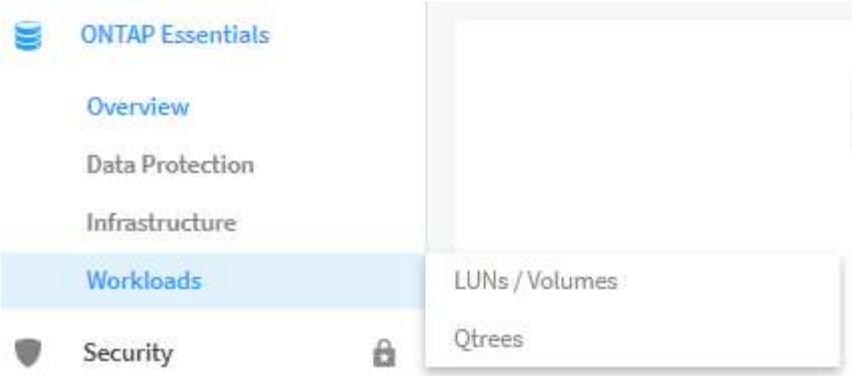
Networking

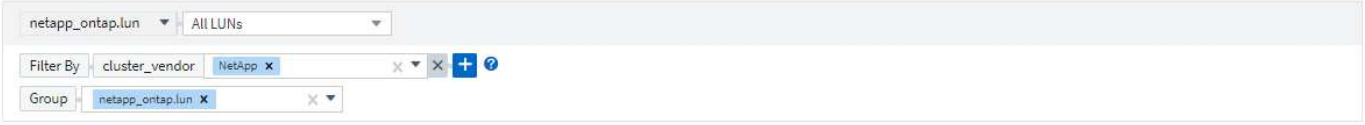
ONTAP Essentials Networking gives you views into your FC, NVME FC, Ethernet, and iSCSI infrastructure. On these pages you can explore things like ports in your clusters and their nodes.



Workloads

View and explore workloads on LUNs/Volumes, NFS or SMB Shares, or Qtrees in your environment.





13 items found

Table Row Grouping	Metrics & Attributes								
netapp_ontap.lun	total_late... ↑	total_iops (IO/s)	total_through...	size (B)	size_used (B)	volume	vserver_name	aggregate_name	node
/vol/ste/ste	0.00	0.00	0.00	53,694,627,840....	0.00	ste	vs_test	umeng_aff300...	u
/vol/kubebug/kubebuglun1	0.00	0.00	0.00	85,905,637,376....	1,489,985,536.00	kubebug	vs_test	umeng_aff300...	u
/vol/trident_pvc_3ef5a87c_4149_44e8_8113...	0.00	0.00	0.00	1,073,741,824.00	0.00	trident_pvc_3e...	vs_test	umeng_aff300...	u
/vol/trident_pvc_0bf4ffd4_3f11_4d63_aa01_...	0.00	0.00	0.00	1,073,741,824.00	0.00	trident_pvc_0b...	vs_test	umeng_aff300...	u
/vol/NSLM_VOL_LUN_1597772263794/matts...	0.00	0.00	0.00	1,073,741,824.00	0.00	NSLM_VOL_LU...	VMware_test	aggr_data_01_...	rt
/vol/mattlun12345/mattlun12345	0.00	0.00	0.00	1,073,741,824.00	0.00	mattlun12345	VMware_test	aggr_data_01_...	rt
/vol/kubebug1/kubebuglun2	0.00	0.00	0.00	85,904,826,368....	0.00	kubebug1	vs_test	umeng_aff300...	u
/vol/trident_pvc_d66d7f51_a623_4fc3_8cda...	0.00	0.00	0.00	1,073,741,824.00	0.00	trident_pvc_d6...	vs_test	umeng_aff300...	u
/vol/Rah/Rah	0.00	0.00	0.00	57,576,960.00	0.00	Rah	vs_test	umeng_aff300...	u
/vol/chap_test_lun_vol/chap_test_lun	0.00	0.00	0.00	107,374,182,40...	0.00	chap_test_lun_...	VMware_test	aggr_data_01_...	rt
/vol/windows_iscsi_example/windows_iscsi...	0.00	0.00	1.04	1,073,741,824.00	10,911,744.00	windows_iscsi...	VMware_test	aggr_data_01_...	rt
/vol/vol_test/lun1	0.04	0.10	0.00	1,073,741,824.00	0.00	vol_test	vs_test	umeng_aff300...	u
/vol/osc_iscsi_vol01/osc_iscsi_vol01	2.11	116.83	2,737,374.33	4,398,046,511,1...	2,535,381,008,3...	osc_iscsi_vol01	osc	umeng_aff300...	u

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